

$$\begin{aligned}
\lambda c &= \mathbf{x}^T A^T \bar{\mathbf{x}} = \mathbf{x}^T (\overline{A \mathbf{x}}) = \mathbf{x}^T (\overline{A} \mathbf{x}) = \mathbf{x}^T (\bar{\lambda} \mathbf{x}) \\
&= \mathbf{x}^T (\bar{\lambda} \bar{\mathbf{x}}) \\
&= \bar{\lambda} \mathbf{x}^T \bar{\mathbf{x}} \\
&= \bar{\lambda} c
\end{aligned}$$

as required. $\square$

The technique in the proof of Theorem 5.5.7 will be used again when we return to complex linear algebra in Section 8.7.

Example 5.5.7

Verify Theorem 5.5.7 for every real, symmetric 2×2 matrix A .

Solution. If $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ we have $c_A(x) = x^2 - (a+c)x + (ac-b^2)$, so the eigenvalues are given by $\lambda = \frac{1}{2}[(a+c) \pm \sqrt{(a+c)^2 - 4(ac-b^2)}]$. But here

$$(a+c)^2 - 4(ac-b^2) = (a-c)^2 + 4b^2 \geq 0$$

for any choice of a , b , and c . Hence, the eigenvalues are real numbers.

Exercises for 5.5

Exercise 5.5.1 By computing the trace, determinant, and rank, show that A and B are *not* similar in each case.

a. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$

c. $A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, B = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$

d. $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

e. $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -3 & 6 & -3 \end{bmatrix}$

f. $A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & -1 & 2 \\ 0 & 3 & -5 \end{bmatrix}, B = \begin{bmatrix} -2 & 1 & 3 \\ 6 & -3 & -9 \\ 0 & 0 & 0 \end{bmatrix}$

Exercise 5.5.2 Show that $\begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 4 & 3 & 0 & 0 \end{bmatrix}$ and

$\begin{bmatrix} 1 & -1 & 3 & 0 \\ -1 & 0 & 1 & 1 \\ 0 & -1 & 4 & 1 \\ 5 & -1 & -1 & -4 \end{bmatrix}$ are *not* similar.

Exercise 5.5.3 If $A \sim B$, show that:

a. $A^T \sim B^T$

b. $A^{-1} \sim B^{-1}$

c. $rA \sim rB$ for r in $\mathbb{R}$

d. $A^n \sim B^n$ for $n \geq 1$

Exercise 5.5.4 In each case, decide whether the matrix A is diagonalizable. If so, find P such that $P^{-1}AP$ is diagonal.

a. $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

b. $\begin{bmatrix} 3 & 0 & 6 \\ 0 & -3 & 0 \\ 5 & 0 & 2 \end{bmatrix}$

c. $\begin{bmatrix} 3 & 1 & 6 \\ 2 & 1 & 0 \\ -1 & 0 & -3 \end{bmatrix}$

d. $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 2 \\ 2 & 3 & 1 \end{bmatrix}$

Exercise 5.5.5 If A is invertible, show that AB is similar to BA for all B .

Exercise 5.5.6 Show that the only matrix similar to a scalar matrix $A = rI$, r in $\mathbb{R}$, is A itself.

Exercise 5.5.7 Let λ be an eigenvalue of A with corresponding eigenvector $\mathbf{x}$. If $B = P^{-1}AP$ is similar to A , show that $P^{-1}\mathbf{x}$ is an eigenvector of B corresponding to λ .

Exercise 5.5.8 If $A \sim B$ and A has any of the following properties, show that B has the same property.

- Idempotent, that is $A^2 = A$.
- Nilpotent, that is $A^k = 0$ for some $k \geq 1$.
- Invertible.

Exercise 5.5.9 Let A denote an $n \times n$ upper triangular matrix.

- If all the main diagonal entries of A are distinct, show that A is diagonalizable.
- If all the main diagonal entries of A are equal, show that A is diagonalizable only if it is *already* diagonal.

- c. Show that $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ is diagonalizable but that $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ is not diagonalizable.

Exercise 5.5.10 Let A be a diagonalizable $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (including multiplicities). Show that:

- $\det A = \lambda_1 \lambda_2 \cdots \lambda_n$
- $\operatorname{tr} A = \lambda_1 + \lambda_2 + \cdots + \lambda_n$

Exercise 5.5.11 Given a polynomial $p(x) = r_0 + r_1x + \cdots + r_nx^n$ and a square matrix A , the matrix $p(A) = r_0I + r_1A + \cdots + r_nA^n$ is called the **evaluation** of $p(x)$ at A . Let $B = P^{-1}AP$. Show that $p(B) = P^{-1}p(A)P$ for all polynomials $p(x)$.

Exercise 5.5.12 Let P be an invertible $n \times n$ matrix. If A is any $n \times n$ matrix, write $T_P(A) = P^{-1}AP$. Verify that:

- $T_P(I) = I$
- $T_P(AB) = T_P(A)T_P(B)$
- $T_P(A+B) = T_P(A) + T_P(B)$
- $T_P(rA) = rT_P(A)$
- $T_P(A^k) = [T_P(A)]^k$ for $k \geq 1$
- If A is invertible, $T_P(A^{-1}) = [T_P(A)]^{-1}$.
- If Q is invertible, $T_Q[T_P(A)] = T_{PQ}(A)$.

Exercise 5.5.13

- Show that two diagonalizable matrices are similar if and only if they have the same eigenvalues with the same multiplicities.
- If A is diagonalizable, show that $A \sim A^T$.
- Show that $A \sim A^T$ if $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Exercise 5.5.14 If A is 2×2 and diagonalizable, show that $C(A) = \{X \mid XA = AX\}$ has dimension 2 or 4. [Hint: If $P^{-1}AP = D$, show that X is in $C(A)$ if and only if $P^{-1}XP$ is in $C(D)$.]

Exercise 5.5.15 If A is diagonalizable and $p(x)$ is a polynomial such that $p(\lambda) = 0$ for all eigenvalues λ of A , show that $p(A) = 0$ (see Example 3.3.9). In particular, show $c_A(A) = 0$. [Remark: $c_A(A) = 0$ for all square matrices A —this is the Cayley-Hamilton theorem, see Theorem 11.1.2.]

Exercise 5.5.16 Let A be $n \times n$ with n distinct real eigenvalues. If $AC = CA$, show that C is diagonalizable.

Exercise 5.5.17 Let $A = \begin{bmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 0 \end{bmatrix}$ and

$$B = \begin{bmatrix} c & a & b \\ a & b & c \\ b & c & a \end{bmatrix}.$$

- Show that $x^3 - (a^2 + b^2 + c^2)x - 2abc$ has real roots by considering A .
- Show that $a^2 + b^2 + c^2 \geq ab + ac + bc$ by considering B .

Exercise 5.5.18 Assume the 2×2 matrix A is similar to an upper triangular matrix. If $\operatorname{tr} A = 0 = \operatorname{tr} A^2$, show that $A^2 = 0$.

Exercise 5.5.19 Show that A is similar to A^T for all 2×2 matrices A . [Hint: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $c = 0$ treat the cases $b = 0$ and $b \neq 0$ separately. If $c \neq 0$, reduce to the case $c = 1$ using Exercise 5.5.12(d).]

Exercise 5.5.20 Refer to Section 3.4 on linear recurrences. Assume that the sequence $x_0, x_1, x_2, \dots$ satisfies

$$x_{n+k} = r_0 x_n + r_1 x_{n+1} + \dots + r_{k-1} x_{n+k-1}$$

for all $n \geq 0$. Define

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ r_0 & r_1 & r_2 & \dots & r_{k-1} \end{bmatrix}, \quad V_n = \begin{bmatrix} x_n \\ x_{n+1} \\ \vdots \\ x_{n+k-1} \end{bmatrix}.$$

Then show that:

- $V_n = A^n V_0$ for all n .
- $c_A(x) = x^k - r_{k-1}x^{k-1} - \dots - r_1x - r_0$
- If λ is an eigenvalue of A , the eigenspace E_λ has dimension 1, and $\mathbf{x} = (1, \lambda, \lambda^2, \dots, \lambda^{k-1})^T$ is an eigenvector. [Hint: Use $c_A(\lambda) = 0$ to show that $E_\lambda = \mathbb{R}\mathbf{x}$.]
- A is diagonalizable if and only if the eigenvalues of A are distinct. [Hint: See part (c) and Theorem 5.5.4.]
- If $\lambda_1, \lambda_2, \dots, \lambda_k$ are distinct real eigenvalues, there exist constants $t_1, t_2, \dots, t_k$ such that $x_n = t_1 \lambda_1^n + \dots + t_k \lambda_k^n$ holds for all n . [Hint: If D is diagonal with $\lambda_1, \lambda_2, \dots, \lambda_k$ as the main diagonal entries, show that $A^n = PD^nP^{-1}$ has entries that are linear combinations of $\lambda_1^n, \lambda_2^n, \dots, \lambda_k^n$.]

Exercise 5.5.21 Suppose A is 2×2 and $A^2 = 0$. If $\operatorname{tr} A \neq 0$ show that $A = 0$.

5.6 Best Approximation and Least Squares

Often an exact solution to a problem in applied mathematics is difficult to obtain. However, it is usually just as useful to find arbitrarily close approximations to a solution. In particular, finding “linear approximations” is a potent technique in applied mathematics. One basic case is the situation where a system of linear equations has no solution, and it is desirable to find a “best approximation” to a solution to the system. In this section best approximations are defined and a method for finding them is described. The result is then applied to “least squares” approximation of data.

Suppose A is an $m \times n$ matrix and $\mathbf{b}$ is a column in $\mathbb{R}^m$, and consider the system

$$A\mathbf{x} = \mathbf{b}$$

of m linear equations in n variables. This need not have a solution. However, given any column $\mathbf{z} \in \mathbb{R}^n$, the distance $\|\mathbf{b} - A\mathbf{z}\|$ is a measure of how far $A\mathbf{z}$ is from $\mathbf{b}$. Hence it is natural to ask whether there is a column $\mathbf{z}$ in $\mathbb{R}^n$ that is as close as possible to a solution in the sense that

$$\|\mathbf{b} - A\mathbf{z}\|$$

Example 6.1.8

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in a vector space V , simplify the expression

$$2(\mathbf{u} + 3\mathbf{w}) - 3(2\mathbf{w} - \mathbf{v}) - 3[2(2\mathbf{u} + \mathbf{v} - 4\mathbf{w}) - 4(\mathbf{u} - 2\mathbf{w})]$$

Solution. The reduction proceeds as though $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ were matrices or variables.

$$\begin{aligned} & 2(\mathbf{u} + 3\mathbf{w}) - 3(2\mathbf{w} - \mathbf{v}) - 3[2(2\mathbf{u} + \mathbf{v} - 4\mathbf{w}) - 4(\mathbf{u} - 2\mathbf{w})] \\ &= 2\mathbf{u} + 6\mathbf{w} - 6\mathbf{w} + 3\mathbf{v} - 3[4\mathbf{u} + 2\mathbf{v} - 8\mathbf{w} - 4\mathbf{u} + 8\mathbf{w}] \\ &= 2\mathbf{u} + 3\mathbf{v} - 3[2\mathbf{v}] \\ &= 2\mathbf{u} + 3\mathbf{v} - 6\mathbf{v} \\ &= 2\mathbf{u} - 3\mathbf{v} \end{aligned}$$

Condition (2) in Theorem 6.1.3 points to another example of a vector space.

Example 6.1.9

A set $\{\mathbf{0}\}$ with one element becomes a vector space if we define

$$\mathbf{0} + \mathbf{0} = \mathbf{0} \quad \text{and} \quad a\mathbf{0} = \mathbf{0} \text{ for all scalars } a.$$

The resulting space is called the **zero vector space** and is denoted $\{\mathbf{0}\}$.

The vector space axioms are easily verified for $\{\mathbf{0}\}$. In any vector space V , Theorem 6.1.3 shows that the zero subspace (consisting of the zero vector of V alone) is a copy of the zero vector space.

Exercises for 6.1

Exercise 6.1.1 Let V denote the set of ordered triples (x, y, z) and define addition in V as in $\mathbb{R}^3$. For each of the following definitions of scalar multiplication, decide whether V is a vector space.

- $a(x, y, z) = (ax, y, az)$
- $a(x, y, z) = (ax, 0, az)$
- $a(x, y, z) = (0, 0, 0)$
- $a(x, y, z) = (2ax, 2ay, 2az)$

Exercise 6.1.2 Are the following sets vector spaces with the indicated operations? If not, why not?

- The set V of nonnegative real numbers; ordinary addition and scalar multiplication.
- The set V of all polynomials of degree ≥ 3 , together with 0; operations of $\mathbf{P}$.
- The set of all polynomials of degree ≤ 3 ; operations of $\mathbf{P}$.
- The set $\{1, x, x^2, \dots\}$; operations of $\mathbf{P}$.

- e. The set V of all 2×2 matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$; operations of $\mathbf{M}_{22}$.
- f. The set V of 2×2 matrices with equal column sums; operations of $\mathbf{M}_{22}$.
- g. The set V of 2×2 matrices with zero determinant; usual matrix operations.
- h. The set V of real numbers; usual operations.
- i. The set V of complex numbers; usual addition and multiplication by a real number.
- j. The set V of all ordered pairs (x, y) with the addition of $\mathbb{R}^2$, but using scalar multiplication $a(x, y) = (ax, -ay)$.
- k. The set V of all ordered pairs (x, y) with the addition of $\mathbb{R}^2$, but using scalar multiplication $a(x, y) = (x, y)$ for all a in $\mathbb{R}$.
- l. The set V of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ with pointwise addition, but scalar multiplication defined by $(af)(x) = f(ax)$.
- m. The set V of all 2×2 matrices whose entries sum to 0; operations of $\mathbf{M}_{22}$.
- n. The set V of all 2×2 matrices with the addition of $\mathbf{M}_{22}$ but scalar multiplication $*$ defined by $a * X = aX^T$.

Exercise 6.1.3 Let V be the set of positive real numbers with vector addition being ordinary multiplication, and scalar multiplication being $a \cdot v = v^a$. Show that V is a vector space.

Exercise 6.1.4 If V is the set of ordered pairs (x, y) of real numbers, show that it is a vector space with addition $(x, y) + (x_1, y_1) = (x + x_1, y + y_1 + 1)$ and scalar multiplication $a(x, y) = (ax, ay + a - 1)$. What is the zero vector in V ?

Exercise 6.1.5 Find $\mathbf{x}$ and $\mathbf{y}$ (in terms of $\mathbf{u}$ and $\mathbf{v}$) such that:

- a. $2\mathbf{x} + \mathbf{y} = \mathbf{u}$
 $5\mathbf{x} + 3\mathbf{y} = \mathbf{v}$
- b. $3\mathbf{x} - 2\mathbf{y} = \mathbf{u}$
 $4\mathbf{x} - 5\mathbf{y} = \mathbf{v}$

Exercise 6.1.6 In each case show that the condition $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \mathbf{0}$ in V implies that $a = b = c = 0$.

- a. $V = \mathbb{R}^4$; $\mathbf{u} = (2, 1, 0, 2)$, $\mathbf{v} = (1, 1, -1, 0)$, $\mathbf{w} = (0, 1, 2, 1)$
- b. $V = \mathbf{M}_{22}$; $\mathbf{u} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
- c. $V = \mathbf{P}$; $\mathbf{u} = x^3 + x$, $\mathbf{v} = x^2 + 1$, $\mathbf{w} = x^3 - x^2 + x + 1$
- d. $V = \mathbf{F}[0, \pi]$; $\mathbf{u} = \sin x$, $\mathbf{v} = \cos x$, $\mathbf{w} = 1$ —the constant function

Exercise 6.1.7 Simplify each of the following.

- a. $3[2(\mathbf{u} - 2\mathbf{v} - \mathbf{w}) + 3(\mathbf{w} - \mathbf{v})] - 7(\mathbf{u} - 3\mathbf{v} - \mathbf{w})$
- b. $4(3\mathbf{u} - \mathbf{v} + \mathbf{w}) - 2[(3\mathbf{u} - 2\mathbf{v}) - 3(\mathbf{v} - \mathbf{w})] + 6(\mathbf{w} - \mathbf{u} - \mathbf{v})$

Exercise 6.1.8 Show that $\mathbf{x} = \mathbf{v}$ is the only solution to the equation $\mathbf{x} + \mathbf{x} = 2\mathbf{v}$ in a vector space V . Cite all axioms used.

Exercise 6.1.9 Show that $-\mathbf{0} = \mathbf{0}$ in any vector space. Cite all axioms used.

Exercise 6.1.10 Show that the zero vector $\mathbf{0}$ is uniquely determined by the property in axiom A4.

Exercise 6.1.11 Given a vector $\mathbf{v}$, show that its negative $-\mathbf{v}$ is uniquely determined by the property in axiom A5.

Exercise 6.1.12

- a. Prove (2) of Theorem 6.1.3. [Hint: Axiom S2.]
- b. Prove that $(-a)\mathbf{v} = -(a\mathbf{v})$ in Theorem 6.1.3 by first computing $(-a)\mathbf{v} + a\mathbf{v}$. Then do it using (4) of Theorem 6.1.3 and axiom S4.
- c. Prove that $a(-\mathbf{v}) = -(a\mathbf{v})$ in Theorem 6.1.3 in two ways, as in part (b).

Exercise 6.1.13 Let $\mathbf{v}, \mathbf{v}_1, \dots, \mathbf{v}_n$ denote vectors in a vector space V and let $a, a_1, \dots, a_n$ denote numbers. Use induction on n to prove each of the following.

- a. $a(\mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_n) = a\mathbf{v}_1 + a\mathbf{v}_2 + \dots + a\mathbf{v}_n$
- b. $(a_1 + a_2 + \dots + a_n)\mathbf{v} = a_1\mathbf{v} + a_2\mathbf{v} + \dots + a_n\mathbf{v}$

Exercise 6.1.14 Verify axioms A2—A5 and S2—S5 for the space $F[a, b]$ of functions on $[a, b]$ (Example 6.1.7).

Exercise 6.1.15 Prove each of the following for vectors $\mathbf{u}$ and $\mathbf{v}$ and scalars a and b .

- If $a\mathbf{v} = \mathbf{0}$, then $a = 0$ or $\mathbf{v} = \mathbf{0}$.
- If $a\mathbf{v} = b\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $a = b$.
- If $a\mathbf{v} = a\mathbf{w}$ and $a \neq 0$, then $\mathbf{v} = \mathbf{w}$.

Exercise 6.1.16 By calculating $(1 + 1)(\mathbf{v} + \mathbf{w})$ in two ways (using axioms S2 and S3), show that axiom A2 follows from the other axioms.

Exercise 6.1.17 Let V be a vector space, and define V^n to be the set of all n -tuples $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ of n vectors $\mathbf{v}_i$, each belonging to V . Define addition and scalar multiplication in V^n as follows:

$$\begin{aligned} (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n) + (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) &= (\mathbf{u}_1 + \mathbf{v}_1, \mathbf{u}_2 + \mathbf{v}_2, \dots, \mathbf{u}_n + \mathbf{v}_n) \\ a(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) &= (a\mathbf{v}_1, a\mathbf{v}_2, \dots, a\mathbf{v}_n) \end{aligned}$$

Show that V^n is a vector space.

Exercise 6.1.18 Let V^n be the vector space of n -tuples from the preceding exercise, written as columns. If A

is an $m \times n$ matrix, and X is in V^n , define AX in V^m by matrix multiplication. More precisely, if

$$A = [a_{ij}] \text{ and } X = \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_n \end{bmatrix}, \text{ let } AX = \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_m \end{bmatrix}$$

where $\mathbf{u}_i = a_{i1}\mathbf{v}_1 + a_{i2}\mathbf{v}_2 + \dots + a_{in}\mathbf{v}_n$ for each i . Prove that:

- $B(AX) = (BA)X$
- $(A + A_1)X = AX + A_1X$
- $A(X + X_1) = AX + AX_1$
- $(kA)X = k(AX) = A(kX)$ if k is any number
- $IX = X$ if I is the $n \times n$ identity matrix
- Let E be an elementary matrix obtained by performing a row operation on the rows of I_n (see Section 2.5). Show that EX is the column resulting from performing that same row operation on the vectors (call them rows) of X . [Hint: Lemma 2.5.1.]

6.2 Subspaces and Spanning Sets

Chapter 5 is essentially about the subspaces of $\mathbb{R}^n$. We now extend this notion.

Definition 6.2 Subspaces of a Vector Space

*If V is a vector space, a nonempty subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space using the addition and scalar multiplication of V .*

Subspaces of $\mathbb{R}^n$ (as defined in Section 5.1) are subspaces in the present sense by Example 6.1.3. Moreover, the defining properties for a subspace of $\mathbb{R}^n$ actually *characterize* subspaces in general.